

Rehand Abdelhady Yassin

Aerospace Engineer

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Aerospace engineering graduate with practical experience in propulsion, CFD, and mechanical design. Skilled in MATLAB, C++, Python, CNC machining, and engineering simulation tools, including ANSYS and SolidWorks. Delivered mathematics instruction at advanced levels. Led a graduation project to design and build a combustion system using kerosene, green hydrogen, and their blends, and I am currently continuing research in combustion systems with a focus on performance analysis and publication.

EDUCATION

- Zewail City of Science and Technology** Sep 2020 – Jun 2025
Bachelor of Science in Aerospace Engineering (Scholarship Recipient) Giza, Egypt

PROFESSIONAL EXPERIENCE

- Zewail City of Science and Technology** 2024 – Present
Undergraduate Researcher — Combustion Systems 6th October, Giza
 - Designed lab-scale combustion system (blower, preheater, combustor, nozzle)
 - Performed CFD simulations using ANSYS Fluent
 - Developed MATLAB models for combustion performance
- Zewail City of Science and Technology** Jul 2025 – Oct 2025
Undergraduate Internship Mentor 6th October, Giza
 - Supervised students in the design and implementation of a CNC foam cutter
 - Provided hands-on training in stepper motor control, mechanical assembly, and system integration
- Egyptian Space Agency (EgSA)** Jul 2025 – Sep 2025
Mechanical Design Engineer-Intern New Cairo City, Cairo
 - Contributed to satellite design, including structural, thermal, and payload subsystems
 - Performed FEA using ANSYS to evaluate stress and structural integrity under launch conditions
- Zewail City Racing Team (Velox Team)** Mar 2023 – Jun 2024
Mechanical Team Lead — Brakes and Wheels (Ever Competition)
 - Designed and analyzed braking and wheel systems using SolidWorks and MATLAB/Simulink for performance and safety
 - Led assembly, component integration, and testing to validate system performance and reliability
- Pulse Jet Engine - Zewail City** May 2023 – Jan 2024
Mechanical Engineering Intern 6th October, Giza
 - Led manufacturing, assembly, and testing of pulse jet engine in workshop environment
 - Designed and analyzed a pulse jet engine using MATLAB and SolidWorks
- EgyptAir Training Academy** Sep 2023
Aircraft Maintenance Intern Cairo, Egypt
 - Worked across Hangars 6000, 7000, and 8000, studying gas turbine engines, gas dynamics, and structural repair processes
 - Gained exposure to aircraft systems, including fuel, ignition, and landing gear, and applied maintenance standards and non-destructive testing (NDT) techniques
- University of Science and Technology at Zewail City** Sep 2021 – May 2022
Junior Teaching Assistant — Mathematics 6th October, Giza
 - Supported students in problem-solving, assignments, and exam preparation
 - Delivered online and in-person tutorials to explain core concepts and improve analytical understanding

PROJECTS

- NACA Airfoil CFD Analysis** Feb 2024 – May 2024
ANSYS Fluent, MATLAB
 - Performed CFD analysis of NACA airfoil to evaluate lift, drag, and pressure distribution
 - Analyzed aerodynamic performance across different angles of attack and validated results
- Satellite Design Project** Jul 2025 – Sep 2025
SolidWorks, ANSYS, MATLAB
 - Designed a full satellite assembly integrating structural, thermal, and subsystem components in SolidWorks
 - Performed FEA in ANSYS and validated kinematic performance using MATLAB

SKILLS

Mechanical

- ANSYS, COMSOL
- Inventor, SOLIDWORKS
- Siemens NX

Programming

- C++
- Python

Electrical & Control

- MATLAB & Simulink
- Proteus, LT Spice
- Arduino IDE

Languages

- Arabic (Native)
- English (Fluent)
- French (Intermediate)

HONORS AND AWARDS

- **3rd Place — Youth Climate Mobility Innovation Hackathon (MENA)** Sep 2025
IOM – UN Migration & YOUNGO
 - Developed Aquavive, an AI-based and energy-efficient water filtration solution for Gaza
 - Collaborated in a multidisciplinary team to design and present a sustainable engineering solution
- **2nd Place — Graduation Project Fair** Jun 2025
Zewail City of Science and Technology
 - Designed a lab-scale combustion chamber using kerosene–hydrogen blends
 - Performed CFD analysis in ANSYS to evaluate performance
- **4th Place — Medhat Ben Zohair Competition** Oct 2022
Greenish Club
 - Developed an entrepreneurship concept supporting rural handicraft communities through sustainable tourism
 - Focused on creating job opportunities and promoting local economic development
- **Global Nominee — NASA Space Apps Challenge** Oct 2022
NASA Space Apps Giza
 - Developed an interactive model simulating the Parker Solar Probe mission from Earth to the solar corona
 - Integrated a QR-based educational game using Unity and G-Develop

LEADERSHIP & ACTIVITIES

- **Ministry of Youth and Sports** Present
Young Negotiator — EGYouth4Climate New Administrative Capital, Egypt
 - Represent Egypt in climate action initiatives and international negotiation simulations
 - Contribute to policy discussions as part of the Innovation and Solutions Committee
- **Be An Ambassador Initiative** Oct 2024
Sustainability Ambassador Intern Cairo, Egypt
 - Developed 32 educational videos on sustainability and climate policy topics
 - Completed intensive training in communication, leadership, and professional development
- **Student Energy** Mar 2024 – Nov 2024
Energy Fellow Online
 - Completed fellowship in energy systems, project development, and energy transition
 - Focused on sustainability and clean energy concepts through the Tier 2 track
- **McKinsey's Forward program** Nov 2023 – Apr 2024
Participant Online
 - Completed training in problem-solving, communication, and digital skills
 - Applied structured problem-solving and data-driven decision-making through projects
- **Zewail City Science Festival (ZCSF)** Jun 2021 – Aug 2023
Organizer / HR / Photographer 6th October, Giza
 - Contributed across three years in event organization, HR coordination, and media coverage
 - Supported planning, team coordination, and documentation of large-scale science events
- **Zewail City Commencement** Jul 2023
Event Organizer 6th October, Giza
 - Selected to support the coordination and organization of the university graduation ceremony
 - Assisted in event logistics, planning, and on-site execution
- **Future Engineers – NASA** May 2019 – Jun 2023
Competitor & Judge Online
 - Participated in NASA competitions, including rover design and space exploration challenges.
 - Served as competition judge, contributing to evaluation and selection processes
- **Admission Team – Zewail City** May 2021-Jan 2022
Admissions Team Member 6th October, Giza
 - Supported student onboarding, registration, and admission process guidance
 - Assisted applicants with procedures, requirements, and inquiries